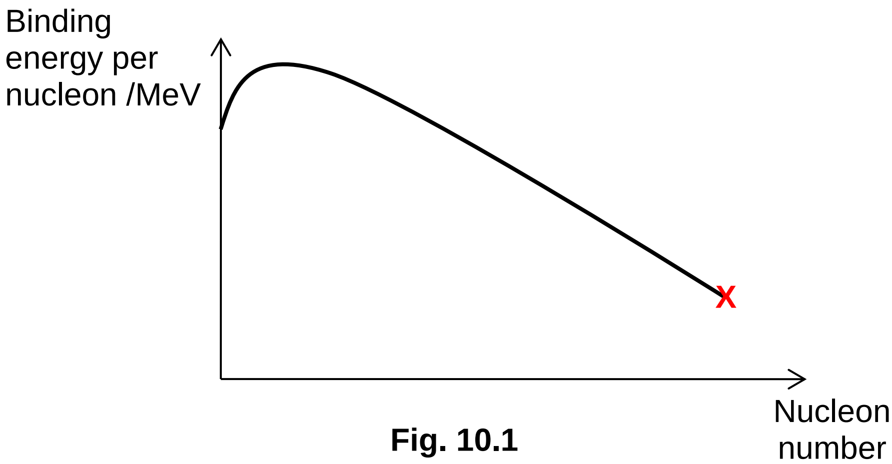
10	**(a)**	**(i)**	The rate of decay is not dependent on physical conditions like temperature, pressure or chemical reactions	**1**
		(ii)	It is impossible to state exactly which nucleus or when a particular nucleus will disintegrate.	**1**
	(b)	**(i)**	$\Delta X/\Delta t$ (Refer to definition of activity)	**1**
		(ii)	$\Delta X/X$ (Number of nuclei that have decayed divided by total number of nuclei initially gives the probability of decay)	**1**
		(iii)	$\dfrac{(\frac{\Delta X}{X})}{\Delta t}$ (Decay constant is the probability per unit time)	**1**
	(c)	**(i)**	$\dfrac{A}{4\pi(0.400)^2}\times(5.0\times10^{-4}) = 250$ $A = 1.0\times10^6\ \text{s}^{-1}$	**1** **1**
		(ii)	135 s is equivalent to 3 half-lives Thus, initial activity = $2^3(1.0\times10^6)$ Using $A = \lambda N$, The initial number of atoms = $2^3(1.0\times10^6)/(0.693/45)$ $= 5.2\times10^8$	**1** **1**
		(iii)	The result obtained is an over-estimation [1]. This is because there may be background radiation present [1]. Hence, the activity due to the source is less than that detected and thus the true initial number of atoms should be less [1].	
		(iv)	Since β-emissions comprises moving electrons, their path of travel can be deflected in a magnetic field.	**1**
	(d)	**(i)**	A large or heavy unstable nucleus splits into two or more smaller nuclei. This process is brought about by neutron bombardment of the nucleus.	**1** **1**
		(ii)	This process may release energy when the binding energy per nucleon increases after the process.	**1**

		(iii)	1.  **Fig. 10.1**	**1**
			2. When the binding energy per nucleon is low, it takes less energy to remove a nucleon from the nucleus.	**1**
		(iv)	Loss in energy = $3.1 \times 10^{-28} \times c^2 = 2.79 \times 10^{-11}$ J So, one uranium-235 nucleus disintegrates with the release of 2.79×10^{-11} J of energy Let *N* be the number of uranium-235 nuclei that release 100 GeV of energy. Then, no. of nuclei required = $100 \times 10^9 \times 1.6 \times 10^{-19} / 2.79 \times 10^{-11} = 570$	**1** **1**